CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (E & C) Examination

May 2014

EC – 406 Digital Image Processing

Date: 24.05.14, Saturday **Time: 01:30 a.m. to 04:30 p.m.** **Maximum Marks: 70**

Instructions:

(1) All questions are compulsory.
(2) Indicate clearly, the option you attempt along with its respective question number.
(3) Figures to right indicate marks
(4)Rough work is done in the last page of main answer book; don't write anything on the question paper.

SECTION-I

Q-1 A Explain in detail the connectivity of pixels. Let V = {0, 1}. Compute Euclidean distance, D8 and D4 distances between two pixels p and q for the image show in figure below. Also compute the length of shortest m-path between p and q. [6]

	0	1	2	3
0	0	1	1	1
1	1	0	0	1
2	1	1	0	1 (q)
3	1(p)	1	1	1

B Answer following questions. [5]

(i) What is meant by illumination and reflectance?
(ii) Differentiate photopic and scotopic vision.
(iii)Compare the 1^{st} order derivative with 2^{nd} order derivative operator.
(iv)Explian the difference between image enhancement and image restoration process.
(v) Define Weber ration.

Q-2 **Attempt any three.**

A Explain the method to calculate the gradient of color image. [4]

B Explain the following edge extraction operator: (i) Sobel (ii) Prewitt (iii) Robert (iv) Laplacian. [4]

C Describe the Wiener filtering and describe is disadvanatge. [4]

D Explain the various high pass frequency domain filter and explain the role of radius in term of image total power. [4]

Q-3 **Attempt any three.**

A Given h(x,y) as follows, discuss the frequency response. **[4]**

0	1/6	0
1/6	1/3	1/6
0	1/6	0

B What is color model? Explain any two color models and their relations. **[4]**

C What are the fundamental steps in image processing? Explain each. **[4]**

D For Given h(x,y) as follow: **[4]**

1	-3	1
-3	9	-3
1	-3	1

Find frequency domain filter H(u,v).

SECTION – II

Q-4 A What is mean by image zooming and shrinking? Explain with example. **[6]**

B How an adaptive median filter can remove the impulse noise? States its advantage too. **[5]**

Q-5 **Attempt any two.**

A Given histograms (i) and (ii), modify the histogram in (i) as given by histogram in (ii). **[6]**

Histogram (i)

Gray level	0	1	2	3	4	5	6	7
Number of pixels	790	1023	850	656	329	245	122	81

Histogram (ii)

Gray level	0	1	2	3	4	5	6	7
Number of pixels	0	0	0	614	819	1230	819	614

B What is Pseudocolor image processing? Explain intensity slicing for color image. **[6]**

C Mention various gray level transformation function and explain any three. **[6]**

Q-6 **Attempt any three.**

A The gary levels in an image range from 10 to 50 . We need to display the image on a device that has a gray level range of 0 to 255. Give transformation which would accomplish this. **[4]**

B Derive mask (size of 3×3) of second derivatives by Laplacian, for enhancement in spatial domain. Prepare the other modified mask of the same. **[4]**

C Explain the Gaussiand noise and Rayleigh noise model with their probability distribution function. **[4]**

D Compare and explain various nonlinear restoration filers. **[4]**